

SADStream: Adaptive Semantic Stream Clustering for Dynamic Topic Detection and Tracking

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Abstract—Social-media streams contain continuously arriving, short, noisy, and rapidly changing text. Traditional clustering methods are poorly suited to this setting because they generally assume a finite dataset and, in the case of K-Means, a predefined number of clusters. Stream-clustering methods such as DenStream address evolving data through incremental micro-clusters and temporal fading, but they were not designed specifically for semantic short-text streams. This research proposes *SADStream*, an online topic detection and tracking framework that combines sentence-level semantic representations, density-based micro-clustering, adaptive forgetting, interpretable topic labeling, and burst-aware event detection. Each topic maintains its own decay rate according to recent activity, while a compact long-term memory stores representations of expired topics so that recurring topics can be distinguished from genuinely new ones. The method will be evaluated on social-media event-detection datasets using clustering quality, event-detection accuracy, detection delay, throughput, and memory consumption. The central research question is whether adaptive semantic stream clustering can improve dynamic event tracking while retaining bounded-memory online operation.

Index Terms—stream clustering, topic detection, event detection, sentence embeddings, adaptive forgetting, topic labeling

I. INTRODUCTION

Social-media platforms generate unbounded streams of short textual messages. Real-world events such as natural disasters, transportation incidents, sporting events, and breaking news may cause new topics to appear suddenly, evolve rapidly, and disappear shortly afterward. Let the incoming stream be

$$S = \{(x_1, t_1), (x_2, t_2), \dots\}, \quad (1)$$

where x_i denotes a social-media post arriving at time t_i . The active topic structure changes over time, such that

$$C_t \neq C_{t+\Delta}. \quad (2)$$

A topic may be created, grow, drift semantically, disappear, or later recur. Social-media topic detection is therefore a non-stationary clustering problem.

Traditional clustering methods are inadequate for this setting because the complete dataset is never available, the number of topics is unknown and changes over time, and older observations should gradually lose importance. Existing stream-clustering algorithms such as DenStream address incremental clustering and temporal fading, while short-text methods such as MStreamF introduce explicit forgetting. Modern sentence embeddings improve semantic similarity for short text, but

semantic representation, adaptive forgetting, recurring-topic recognition, interpretable labeling, and breaking-event detection are rarely handled within one bounded-memory framework.

The main research question is: *Can semantic online clustering with adaptive forgetting improve real-time topic detection and tracking under evolving social-media streams?* SADStream addresses four responsibilities: semantic representation of incoming posts, incremental topic clustering and labeling, temporal adaptation and forgetting, and event detection and tracking.

II. RELATED WORK AND RESEARCH GAP

Stream-clustering algorithms maintain compact summaries instead of storing an entire data stream. DenStream uses potential and outlier micro-clusters together with temporal fading, allowing clusters to appear and disappear without requiring the number of clusters in advance [1].

Short-text stream clustering has been addressed by methods such as MStream and MStreamF [2]. MStreamF is particularly relevant because it includes a forgetting mechanism for removing outdated clusters. However, these approaches primarily rely on lexical or probabilistic text representations.

Modern topic-modeling systems increasingly use contextual embeddings. BERTopic, for example, clusters transformer-based document embeddings and produces interpretable topic representations using class-based TF-IDF (c-TF-IDF) [3]. Recent event-detection work also demonstrates that sentence embeddings can be combined with incremental clustering [5].

Three limitations remain. First, semantic event-detection approaches are often batch- or window-based rather than true bounded-memory stream-clustering systems. Second, traditional stream-clustering algorithms commonly use a fixed global forgetting rate,

$$f(\Delta t) = e^{-\lambda \Delta t}, \quad (3)$$

which assumes that all topics have similar lifetimes. Third, deleting inactive clusters completely makes it difficult to distinguish a genuinely new topic from a previously observed topic that has returned. The proposed research addresses these limitations by integrating semantic representation, online micro-clustering, adaptive forgetting, topic labeling, and compressed topic memory.

III. PROPOSED METHOD

SADStream processes each post through semantic representation, online micro-clustering, temporal adaptation, and event detection. Each stage operates incrementally so that processing and storage remain bounded with respect to stream length.

A. Semantic Representation

Each incoming post x_t is converted into a dense semantic embedding,

$$e_t = \text{Encoder}(x_t), \quad \hat{e}_t = \frac{e_t}{\|e_t\|_2}. \quad (4)$$

Because event-specific information such as hashtags and named entities may remain important, the assignment procedure uses the hybrid similarity function

$$S(x, C_k) = \alpha S_{\text{sem}} + \beta S_{\text{lex}} + \gamma S_{\text{ent}} + \delta S_{\text{hash}}, \quad (5)$$

where the four terms measure semantic, lexical, named-entity, and hashtag similarity, respectively. Their contribution will be measured through ablation experiments.

B. Online Topic Clustering

The system maintains a set of micro-clusters

$$M_t = \{MC_1, \dots, MC_m\}. \quad (6)$$

Each micro-cluster stores compact statistics,

$$MC_k = (\mu_k, W_k, N_k, T_k, K_k, E_k, H_k), \quad (7)$$

where μ_k is the semantic centroid, W_k is temporal weight, N_k is the number of assigned posts, T_k contains temporal metadata, and K_k , E_k , and H_k store representative keywords, named entities, and hashtags.

For a new post, the closest compatible cluster is selected as

$$MC^* = \arg \max_k S(x_t, MC_k). \quad (8)$$

If the maximum similarity exceeds an assignment threshold θ_A , the observation is absorbed; otherwise, a new candidate micro-cluster is created. Its weight and centroid are updated by

$$W'_k = W_k f_k(\Delta t) + 1, \quad (9)$$

$$\mu'_k = \frac{W_k f_k(\Delta t) \mu_k + e_t}{W'_k}. \quad (10)$$

A candidate is promoted to an active topic when $W_k > \theta_p$. Consequently, the number of active topics changes dynamically.

C. Topic Formation and Labeling

Each active micro-cluster is a candidate topic. Closely related micro-clusters may optionally be grouped into a higher-level topic when their centroids and auxiliary features are sufficiently similar. For two micro-clusters, define

$$S_{\text{topic}}(MC_i, MC_j) = \alpha_c \cos(\mu_i, \mu_j) + \beta_c J(K_i, K_j) + \gamma_c J(E_i, E_j) + \delta_c J(H_i, H_j), \quad (11)$$

where $J(\cdot, \cdot)$ is Jaccard similarity. Clusters whose similarity exceeds a merge threshold θ_M are treated as semantic variations of the same topic. This optional aggregation retains incremental processing while allowing a real-world event to be represented by several local micro-clusters.

Every topic receives a persistent identifier, such as `Topic-017`, and a human-readable label. The identifier supports lifecycle tracking; the label summarizes the topic for users. Label generation uses

$$R_k = (K_k, E_k, H_k), \quad (12)$$

where keywords are ranked using c-TF-IDF, TF-IDF, or another lightweight term-weighting method and combined with salient entities and frequent hashtags. The highest-ranked non-redundant elements form a short descriptive phrase. For example, `Topic-017` may receive the label “Tokyo earthquake.”

Labels are updated incrementally as posts enter a cluster. To avoid unstable names, a label changes only when newly dominant terms persist for a minimum number of observations or update intervals. The persistent topic identifier remains unchanged when its descriptive label evolves. If an inactive topic recurs, similarity against its archived centroid, keywords, entities, and hashtags recovers the previous identifier where appropriate, while the label is refreshed from recent posts.

D. Temporal Adaptation and Forgetting

The primary methodological contribution is cluster-specific adaptive forgetting. Conventional temporal fading uses one decay parameter, whereas SADStream assigns each topic its own rate:

$$f_k(\Delta t) = e^{-\lambda_k(t) \Delta t}. \quad (13)$$

A preliminary formulation is

$$\lambda_k(t) = \frac{\lambda_0}{1 + \eta B_k(t) + \rho A_k(t)}, \quad (14)$$

where $A_k(t)$ represents recent activity and $B_k(t)$ represents burstiness. Active or bursty topics therefore decay more slowly, while inactive topics disappear more quickly.

When $W_k < \theta_{\text{inactive}}$, the topic is removed from active memory. Rather than deleting it completely, the system stores the compressed prototype

$$L_k = (\mu_k, K_k, E_k, H_k, t_{\text{start}}, t_{\text{end}}). \quad (15)$$

When a new candidate appears, it is compared with archived prototypes. If

$$\text{sim}(C_{\text{new}}, L_j) > \theta_R, \quad (16)$$

the cluster is classified as recurring rather than completely new.

E. Event Detection and Tracking

Cluster size alone does not determine whether a topic is breaking. SADStream therefore measures changes in recent activity. For topic k ,

$$A_k(t) = \sum_{x_i \in C_k} e^{-\lambda_k(t-t_i)}. \quad (17)$$

TABLE I
PROPOSED EXPERIMENTAL BASELINES

Model	Purpose
TF-IDF + incremental/window clustering	Lexical baseline
Sentence embedding + incremental clustering	Semantic baseline
DenStream	General stream baseline
MStreamF	Short-text forgetting baseline
SADStream with fixed decay	Adaptive-decay ablation
Full SADStream	Proposed model

Burstiness is estimated as

$$B_k(t) = \frac{A_k(t) - \mu_{A,k}}{\sigma_{A,k} + \epsilon}. \quad (18)$$

The breaking-event score is

$$\text{Score}_k = w_1 B_k + w_2 N_k + w_3 D_k + w_4 U_k, \quad (19)$$

where B_k is burstiness, N_k is novelty, D_k is semantic density, and U_k is user or source diversity. An alert is generated when $\text{Score}_k > \theta_B$.

Each topic receives a persistent identifier and a dynamically updated human-readable label. It can occupy one of five lifecycle states: birth, active, decaying, dead, or recurring.

IV. RESEARCH QUESTIONS

The study focuses on three primary research questions:

- RQ1: Does semantic representation improve incremental event clustering compared with lexical representations such as TF-IDF?
- RQ2: Does cluster-specific adaptive forgetting improve dynamic topic tracking compared with fixed exponential decay?
- RQ3: Does retaining compressed representations of inactive topics improve recurring-topic recognition without substantially increasing memory usage?

A secondary research question examines whether burst-aware ranking detects emerging events earlier than ranking clusters using message volume alone. Topic-label coherence and stability will also be examined as interpretability outcomes.

V. EXPERIMENTAL EVALUATION

A. Datasets

The primary benchmarks are Event2012 and Event2018, the latter described in [4]. Posts will be processed in chronological order to preserve the streaming setting. A Vietnamese social-media dataset may be added as a language-transfer experiment if suitable data and annotation resources are available.

B. Baselines

Table I summarizes the minimum baseline set. The most important controlled comparison is SADStream with fixed decay against SADStream with adaptive decay.

C. Evaluation Metrics

Clustering quality will be measured using the adjusted Rand index (ARI), adjusted or normalized mutual information (AMI/NMI), and B^3F_1 . Event-detection performance will use precision, recall, and F_1 . Detection delay for event E is

$$\text{Delay}(E) = t_{\text{detected}}(E) - t_{\text{start}}(E). \quad (20)$$

Streaming performance will additionally measure posts processed per second, per-post latency, peak memory consumption, and the number of active micro-clusters.

Human-readable labels will be assessed using topic coherence, label stability across update intervals, and agreement with available event descriptions. If reference labels are unavailable, a small blinded human evaluation will compare whether labels accurately and concisely describe sampled cluster posts.

D. Ablation Study

The minimum ablation study compares the full model with variants that remove adaptive forgetting, long-term memory, semantic representation, and topic-level merging or auxiliary labeling signals. These experiments determine whether each claimed component contributes materially to clustering, tracking, recurrence detection, or interpretability.

VI. EXPECTED CONTRIBUTIONS

The study targets four contributions. First, it proposes a bounded-memory online clustering framework using contextual semantic representations without requiring the number of topics in advance. Second, it introduces a topic-specific decay mechanism that supports different temporal lifetimes. Third, it maintains a compact semantic archive that distinguishes new topics from recurring topics. Fourth, it adds stable, incrementally updated labels that separate persistent computational identity from human-readable topic descriptions.

The central hypothesis is that adaptive semantic stream clustering can track emerging and disappearing social-media topics more effectively than fixed-decay streaming methods while retaining bounded-memory operation.

VII. MINIMUM IMPLEMENTATION SCOPE

The minimum implementation comprises a sentence encoder, DenStream-style micro-clusters, adaptive decay, keyword/entity/hashtag summaries, stable topic labels, a compressed topic archive, and a burst score. It does not require training a language model from scratch, supervised event classification, misinformation detection, sentiment analysis, full social-network graph modeling, sophisticated merge/split detection, or production-scale distributed deployment.

The minimally successful result is a chronological stream experiment demonstrating whether adaptive decay improves topic clustering or tracking quality over fixed decay while maintaining acceptable latency and memory use. Topic labels should additionally be more coherent and stable than labels produced from unweighted frequent terms.

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